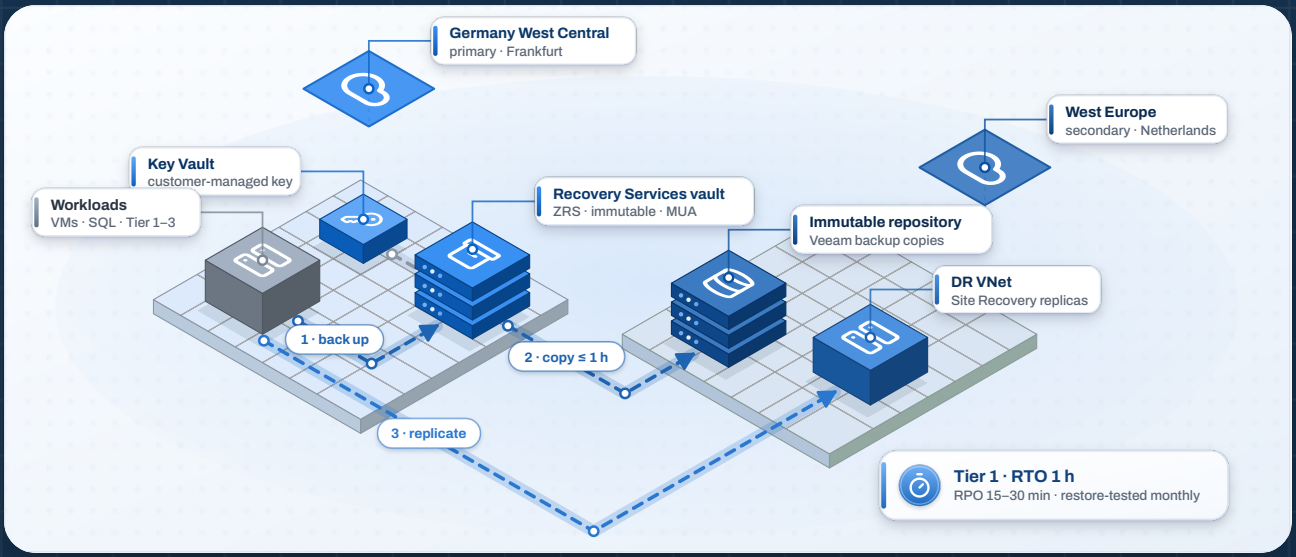
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-AZ-005

Designing Multi-Region Backup and Disaster Recovery in Azure

Protect VMs and SQL across two EU regions with native Azure Backup (portal and PowerShell), Veeam Backup for Microsoft Azure and Azure Site Recovery, then prove it with restore testing.

Microsoft Azure | Backup & Disaster Recovery | Compliance



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Define RPO and RTO and tell backup, replication and disaster recovery apart.
- Choose storage redundancy (LRS, ZRS, GRS) and the right vault type for each workload.
- Harden backups with soft delete, immutability, customer-managed keys and private endpoints.
- Build native Azure Backup in the portal and with a PowerShell template, including SQL in VM.
- Add Veeam Backup for Microsoft Azure for cross-region copies and Site Recovery for Tier 1.
- Run restore tests and map the design to GDPR, NIS2, KRITIS and BSI C5 expectations.

TIME	DIFFICULTY	MINIMUM ROLE	RISK
1–2 days build + monthly tests	Advanced	Backup Contributor + Site Recovery Contributor (RG)	Low to build; high if never tested

CHAPTER 1

Concepts You Need First

Every organization says it "has backups". Far fewer can say how much data they would lose, how long they would be down, and whether an attacker who owns their admin accounts could delete those backups first. This SOP designs a **two-region** protection scheme in Azure for a regulated industrial (chemical-sector) operator in Germany, with production in **Germany West Central (Frankfurt)** and off-site copies and failover in **West Europe (Netherlands)**. The ideas apply to any organization.

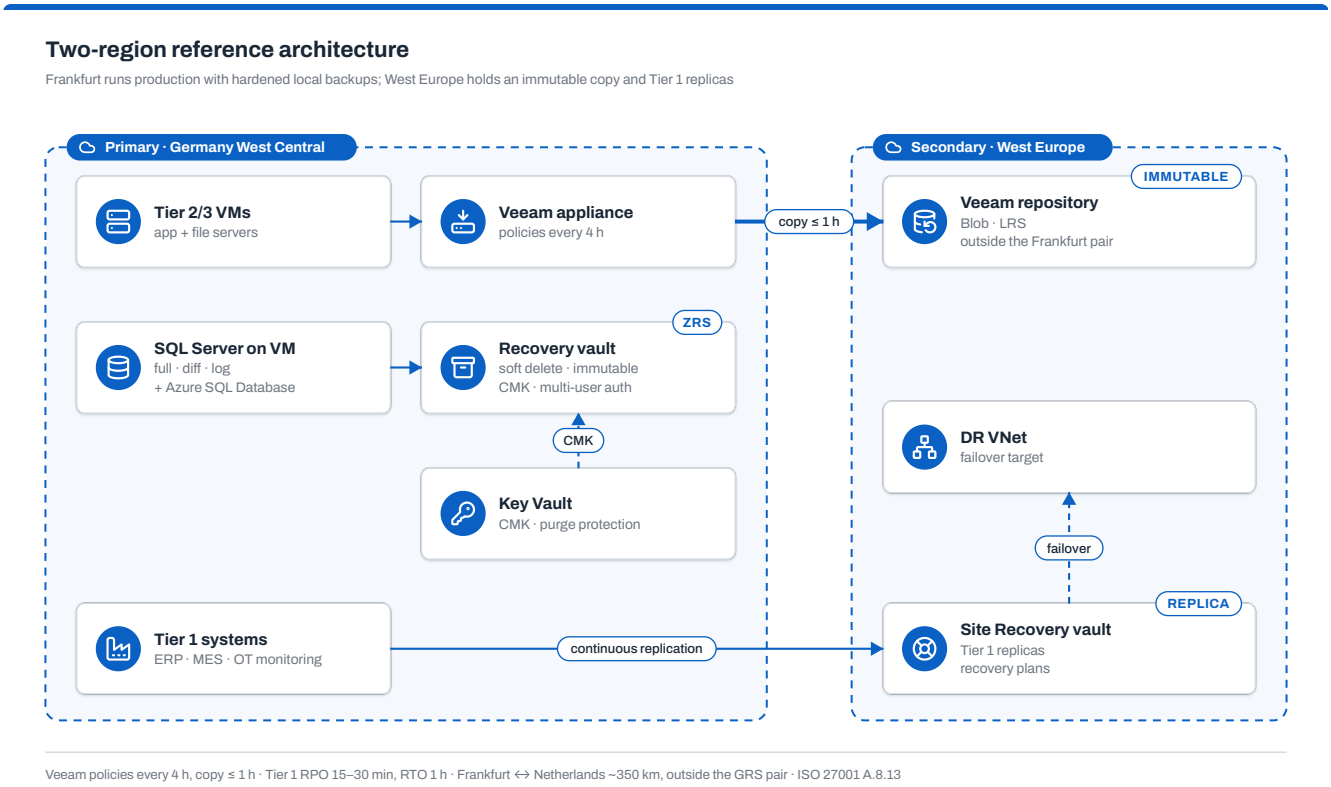


Figure 1. Every arrow in the reference design is one of the methods in this SOP, so a missing arrow is a missing method.

1.1 RPO and RTO

Two numbers drive every decision in this document:

- **Recovery Point Objective (RPO)**: the maximum amount of data, measured in time, you can afford to lose. Backups every 4 hours mean up to 4 hours of work may be gone.
- **Recovery Time Objective (RTO)**: the maximum time a service may be unavailable, from the incident until users are working again. It includes detecting the problem, deciding to restore, the restore itself and the checks afterwards.

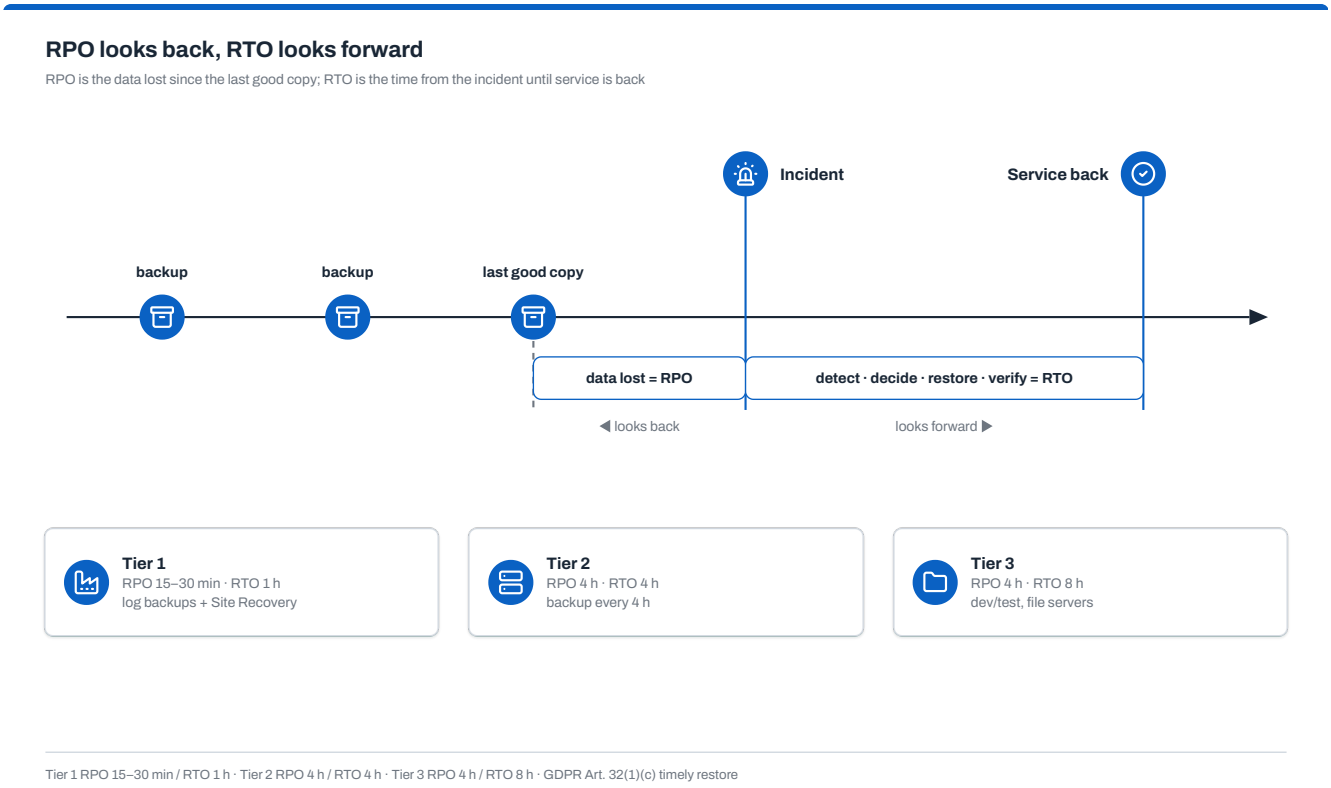


Figure 2. RPO is set by how often you copy and RTO by how fast you can restore, so each tier needs both numbers before you choose a method.

KEY CONCEPT · INSURANCE, NOT A GUARANTEE

RPO and RTO are business decisions expressed in minutes. A shorter RPO needs more frequent copies; a shorter RTO needs copies that are already "warm" in the right place. Both cost money, so not every system gets the same treatment. That is why workloads are put into **tiers**.

1.2 Backup, replication and disaster recovery

Technique	What it does	Protects against
Backup	Point-in-time copies kept for days or weeks. You can go *back* to yesterday.	Deletion, corruption, ransomware, bad changes.
Replication	Continuously copies changes to another place. The copy is almost current.	Hardware or region failure. **Not** corruption: bad data replicates too.
Disaster recovery	The plan, runbooks and tooling that use backups and replicas to restore service elsewhere.	Loss of a site or region, major incidents.

WATCH OUT · REPLICATION IS NOT BACKUP

If ransomware encrypts a database, replication faithfully copies the encrypted data to the second region within seconds. You need backups with history *and* replication for speed. This design uses both.

1.3 Storage redundancy and region pairs

Option	Copies kept	Survives
LRS	3 copies in one datacenter	Disk or rack failure.
ZRS	3 copies across availability zones in one region	Loss of one datacenter (zone).
GRS	LRS in the primary region + asynchronous copy to the *paired* region	Loss of a whole region, with some lag.

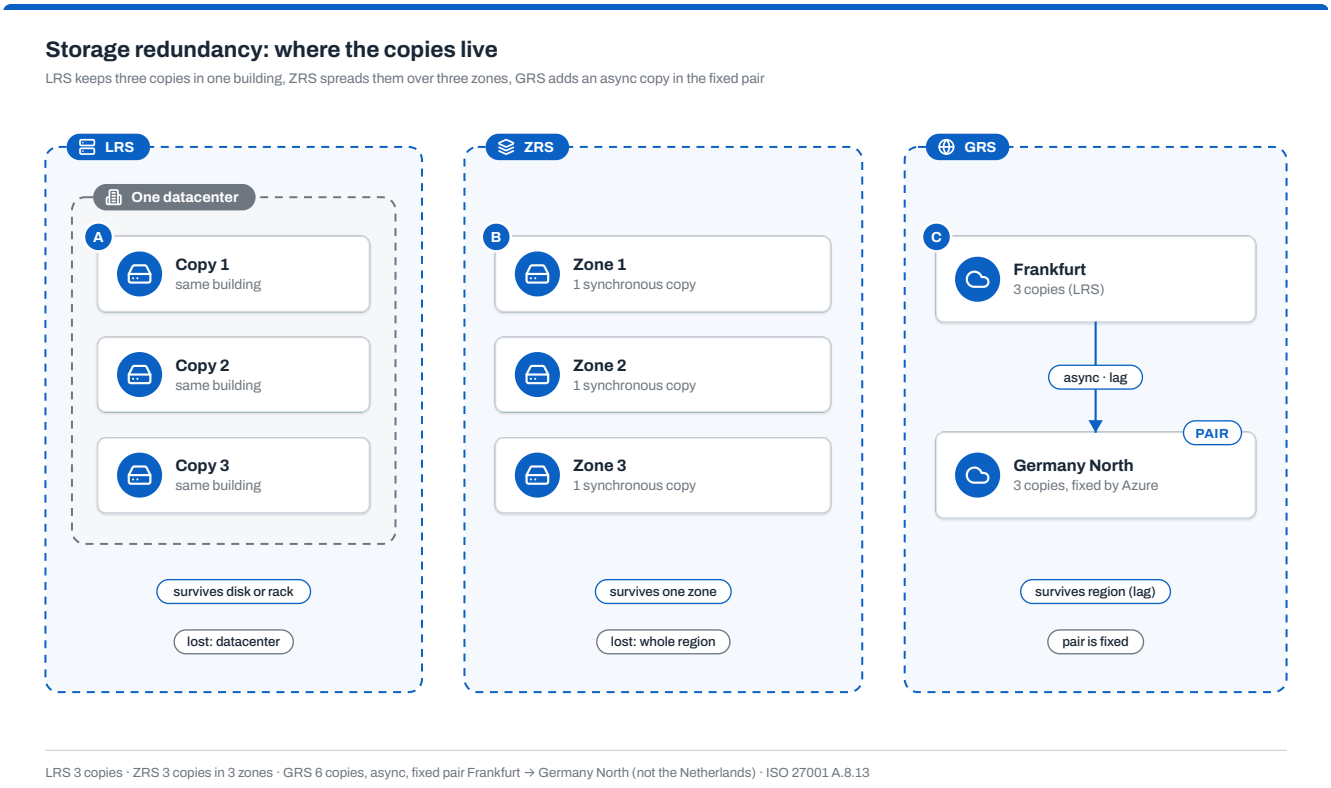


Figure 3. GRS copies to a fixed pair you cannot choose, so a copy in the Netherlands needs Veeam or Site Recovery, not a storage setting.

GRS always replicates to Azure's fixed **paired region**. Germany West Central is paired with **Germany North** (a restricted-access region), not with West Europe. A vault in Frankfurt set to GRS therefore keeps its geo-copy in Germany North, and Cross Region Restore restores there. To hold copies in the *Netherlands*, as this design requires, you need a mechanism that copies to a region you choose: Veeam backup copy jobs (Method C) or

Azure Site Recovery (Method D). That is why the reference design uses ZRS locally and lets those tools handle the cross-region leg.

1.4 Recovery Services vaults and Backup vaults

Azure has two vault types. A **Recovery Services vault** protects Azure VMs, SQL Server and SAP HANA inside VMs, Azure Files and on-premises servers (MARS agent), and also hosts Site Recovery. A **Backup vault** (the newer Data Protection platform) protects Azure Disks, Blobs, Azure Database for PostgreSQL and AKS. Both are managed from **Backup center** (Business Continuity Center) in the portal. This SOP uses Recovery Services vaults for VMs, SQL and Site Recovery.

1.5 Protecting the backups themselves

Modern ransomware operators look for backups first. Four vault features stop them:

- **Soft delete**: deleted backup data is kept for 14 days (extendable up to 180) and can be undeleted. Enhanced soft delete can be made *always-on*, which cannot be turned off.
- **Immutability**: an immutable vault blocks operations that would destroy recovery points early, such as reducing retention or stopping protection with data deletion. *Unlocked* can still be reversed; *Locked* cannot.
- **Multi-user authorization (MUA)**: critical operations require permission on a separate **Resource Guard** owned by the security team, so one compromised backup admin cannot disable protection.
- **Customer-managed keys (CMK)**: backup data is encrypted with AES-256 using a key you control in Azure Key Vault (with soft delete and purge protection). You can revoke access, and key usage is logged.



Figure 4. Each layer stops a different attack, so removing any one of them leaves a path to the recovery points; each is configured in Method A or B.

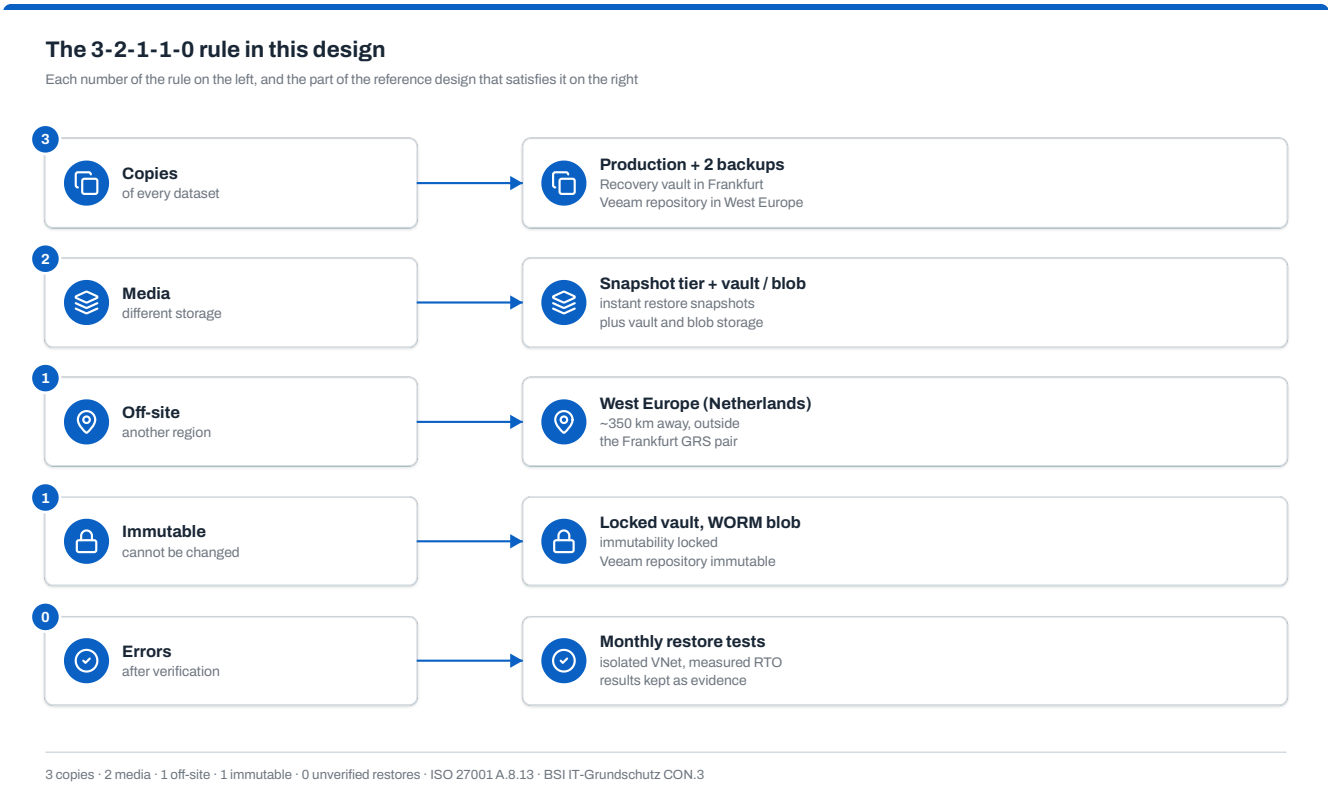


Figure 5. The rule is only met when the last digit is proven: zero errors means a restore test, not a green backup job.

1.6 Consistency and SQL backup types

A **crash-consistent** copy is like pulling the power cord: disks are intact but in-memory writes may be lost. An **application-consistent** copy asks the application to flush and pause writes first (Volume Shadow Copy Service on Windows, pre/post scripts on Linux), so databases restart cleanly. Azure Backup and Veeam both take application-consistent snapshots where possible.

Databases need more than VM snapshots, because a 4-hour RPO is too coarse for transaction systems. SQL Server uses three backup types together:

Type	Contains	Typical schedule here
Full	The whole database at a point in time.	Daily (Azure Backup also allows weekly).
Differential	Everything changed since the last full.	Daily, used with weekly fulls to cut storage.
Transaction log	Every committed transaction since the last log backup; enables **point-in-time restore**.	Every 15–30 minutes. This sets the Tier 1 RPO.

Azure SQL Database (PaaS) does this automatically: weekly fulls, differentials every 12–24 hours and log backups roughly every 10 minutes, with point-in-time restore for 1–35 days, optional long-term retention for years, and geo-restore into any region from geo-redundant backup storage.

1.7 Choosing a method

A · Native Azure Backup (portal)
Vault, policy and protection through Backup center.
BEST FOR: FIRST BUILD, SMALL ESTATES

B · PowerShell template
Vault hardening, enhanced 4-hour policy, VM and SQL protection as code.
BEST FOR: REPEATABLE, AUDITED BUILDS

C · Veeam Backup for Azure
Third-party policies, cross-region copies to any region, portable formats.
BEST FOR: MULTI-REGION COPIES, VEEAM SHOPS

D · Azure Site Recovery
Continuous replication and orchestrated failover.
BEST FOR: TIER 1, MINUTES OF RTO

E · Restore testing
Monthly proof that A–D actually work.
BEST FOR: EVERYONE, ALWAYS

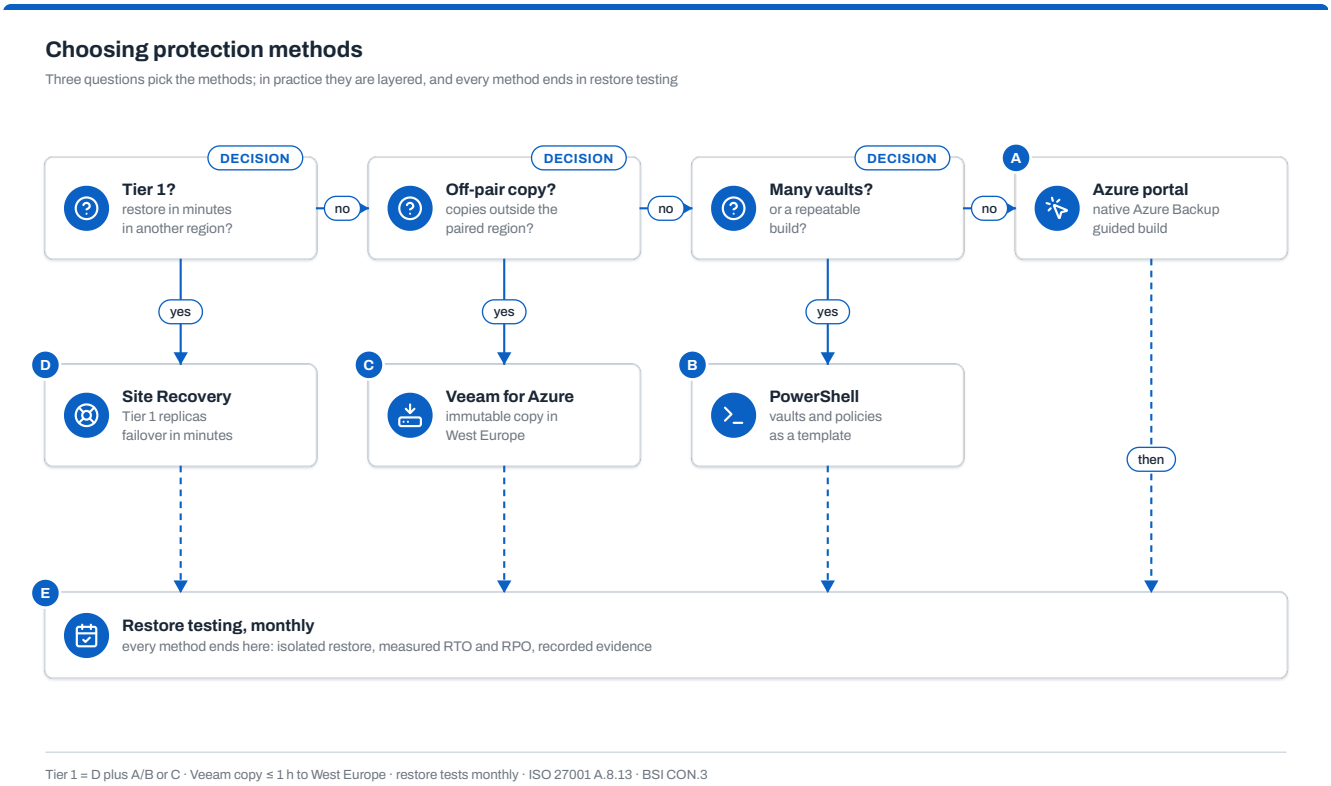


Figure 6. The methods are layered rather than exclusive: Tier 1 systems use D plus A/B or C, and every method ends in restore testing.

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Backups are a prime target, so **separation of duties** matters more here than almost anywhere. The people who run daily backups should not be able to delete vaults, change encryption keys or disable the Resource Guard.

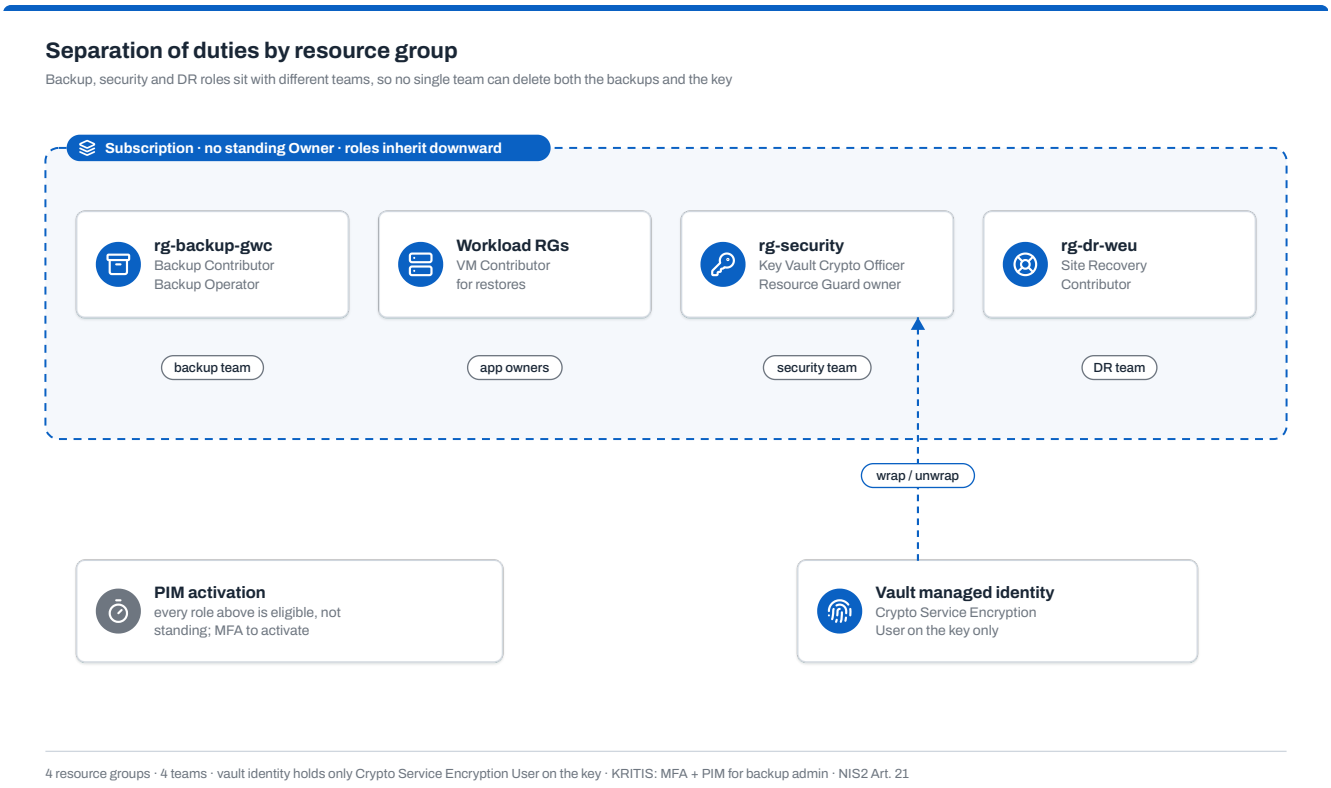


Figure 7. Splitting roles across teams means a single stolen account cannot delete both the backups and the key that decrypts them.

Task	Least-privilege built-in role	Scope and notes
Create and configure vaults and policies	Backup Contributor	Backup RG. Cannot delete the vault or grant access to others.
Enable protection on a VM	Backup Contributor + Virtual Machine Contributor	VM Contributor on the workload RG (installs the backup extension).
Daily operations: run jobs, restore	Backup Operator	Backup RG. Cannot remove backups or change policies. Restores also need VM Contributor and network-join on the target RG.

Read-only monitoring and reports	Backup Reader + Log Analytics Reader	Backup RG and workspace.
Manage CMK keys	Key Vault Crypto Officer	Security team, Key Vault in `rg-security`.
Vault uses the key	Key Vault Crypto Service Encryption User	Granted to the vault's managed identity on the key only.
Approve critical operations (MUA)	Contributor on the Resource Guard (time-bound)	Resource Guard owned by security, ideally in another subscription.
Configure replication and failover	Site Recovery Contributor (Site Recovery Operator for failover only)	DR RG in West Europe, plus VM Contributor and Network Contributor on target RGs.
Veeam service account	Custom role per Veeam's permission list	Entra app registration scoped to the protected RGs; avoid subscription Owner.

GOOD PRACTICE · JUST-IN-TIME AND BREAK-GLASS

Make every backup and DR role *eligible* in Privileged Identity Management with MFA and approval for Backup Contributor and Resource Guard access. Keep two break-glass accounts excluded from Conditional Access lockouts, documented in the DR runbook, so a recovery is never blocked by the identity system it is recovering.

2.2 Regions, tiers and policies

Item	Primary · Germany West Central	Secondary · West Europe
Role	Production, local backups, Key Vault	Off-site copies, Site Recovery target, DR VNet
Why this region	Data residency in Germany, broad service catalogue, DE-CIX connectivity	Mature EU region ~350 km away, AMS-IX connectivity, GDPR scope
Redundancy	ZRS vault (LRS acceptable for Tier 3)	LRS repository / vault (it is already the second region)
Retention	14 days (2 weeks)	35 days (5 weeks)

Workload	Frequency	Tool
VMs (all tiers)	Every 4 hours, application-consistent	Azure Backup enhanced policy and/or Veeam
SQL Server on VM	Full daily, differential optional, log every 15–30 min	Azure Backup (SQL in VM) and/or Veeam

Azure SQL Database	Automatic; PITR 35 days, geo-redundant backup storage	Platform, plus LTR where needed
Tier 1 VMs	Continuous replication	Azure Site Recovery to West Europe
Configuration	On change	Infrastructure as code in Git

> **NOTE · CORRECTING A COMMON ASSUMPTION**
>
> Older designs, including the source notes for this SOP, assumed native Azure VM backup runs only once a day. That is no longer true: the **Enhanced** VM backup policy supports schedules every 4, 6, 8 or 12 hours. The real reason to add Veeam here is the cross-region copy to a non-paired region and portable backup formats, not frequency.

2.3 Prerequisites checklist

- [ ] Approved change record and a `ChangeRef` tag value; workload inventory with tier, owner, RPO and RTO.
- [ ] Hub VNets in both regions, peered, with Azure Firewall and private DNS zones for `privatelink` endpoints.
- [ ] Key Vault (soft delete + purge protection) and a backup encryption key; Log Analytics workspace.
- [ ] Resource Guard in a security-owned subscription; Veeam licensing if Method C is used.
- [ ] A documented decision on immutability: start *Unlocked*, move to *Locked* after a successful test cycle.

CHAPTER 3

Method A · Native Azure Backup in the Portal

All steps start in **Backup center** (also reachable as *Business Continuity Center*) after activating your PIM role.

STEP 1 Create and harden the vault

1. **Backup center › + Vault › Recovery Services vault.** Resource group `rg-backup-gwc`, name `rsv-prod-gwc-01`, region Germany West Central, tags.
2. Before protecting anything, open **Properties › Backup Configuration** and set storage redundancy to **Zone-redundant**. It cannot be changed once items are protected.
3. **Properties › Security settings:** confirm soft delete is on, set retention (e.g. 28 days), and consider *always-on*. Enable **immutability** in *Unlocked* state.
4. **Properties › Encryption settings:** enable a system-assigned identity, grant it *Key Vault Crypto Service Encryption User* on the key, then select the key.
5. **Properties › Multi-User Authorization:** link the security team's Resource Guard.
6. **Networking:** deny public access and add private endpoints for the *Backup* sub-resource in the hub VNet.

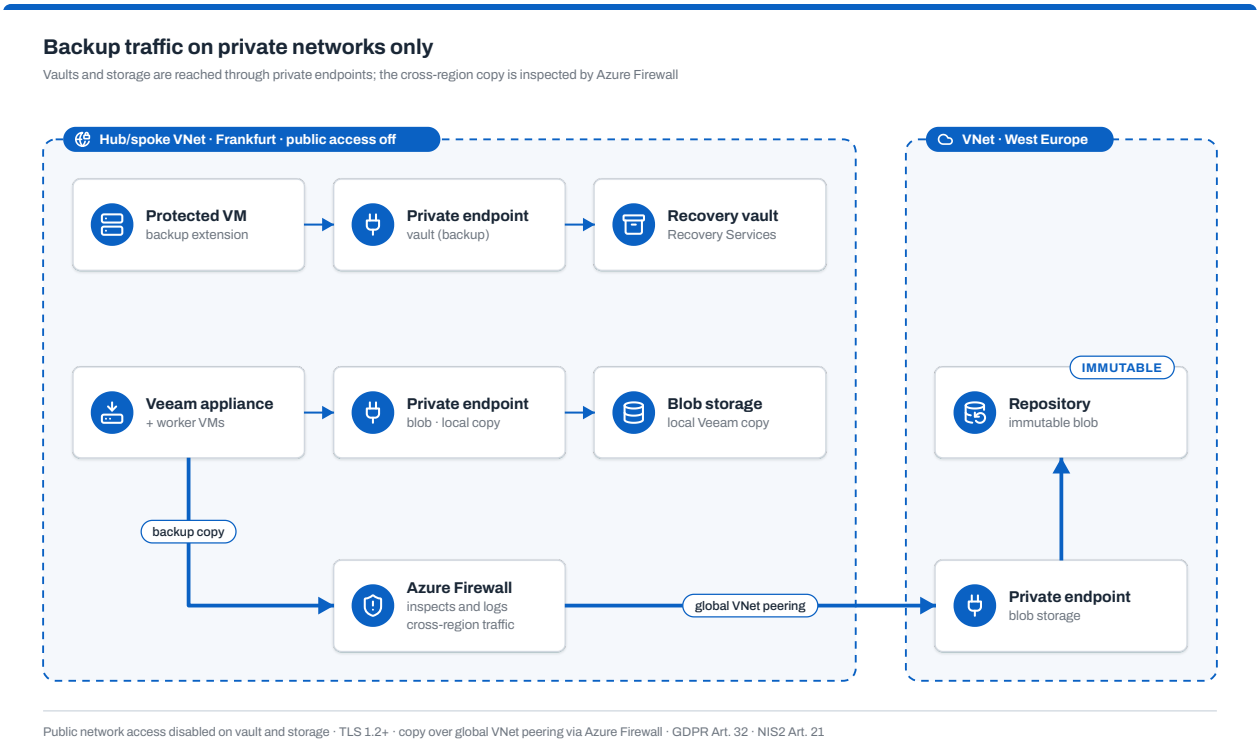


Figure 8. With public access disabled, every backup path needs a private endpoint; if one is missing, the job fails rather than falling back to the internet.

STEP 2 Create the 4-hour enhanced policy

1. Vault › Backup policies › + Add › Azure Virtual Machine, choose **Enhanced**.
2. Frequency **Hourly**, every **4 hours**, start 00:00, duration 24 hours (time zone UTC).
3. Instant restore: keep snapshots 7 days (fast restores from the snapshot tier); daily retention 14 days; weekly/monthly points only if the business requires them.
4. Name it `pol-vm-4h-14d` and create it.

STEP 3 Protect VMs and SQL databases

1. Backup center › + Backup › Azure Virtual Machine, pick the vault and policy, select the VMs, **Enable backup**. Then run **Backup now** once to seed the first recovery point.

2. For SQL Server in a VM: **+ Backup › SQL in Azure VM › Start discovery**, select the VM, then **Configure backup**. Choose databases (enable *auto-protect* on the instance so new databases are covered) and a policy with daily full and 15-minute log backups.
3. For Azure SQL Database, open **SQL server › Backups › Retention policies**, set PITR to 35 days and add LTR where needed; choose geo-redundant backup storage on the database.

WATCH OUT · TWO BACKUP ENGINES ON ONE DATABASE

Do not let Azure Backup *and* another tool (Veeam, SQL Agent jobs) both take SQL **log** backups of the same database: each breaks the other's log chain and point-in-time restore fails. Pick one owner for log backups; the other may take copy-only fulls.

STEP 4 Monitor and alert

1. Send vault diagnostics (resource-specific tables) to Log Analytics; enable **Backup center › Backup reports**.
2. Use built-in Azure Monitor alerts for job failures and security events (e.g. soft delete disabled, immutability changed) and route them to the on-call action group and Microsoft Sentinel.

CHAPTER 4

Method B · PowerShell Base Template

The template creates a hardened Recovery Services vault, sets redundancy, soft delete, immutability and optional CMK, builds an enhanced 4-hour policy and protects a list of VMs. It writes a JSON evidence file for the change record and supports `-WhatIf`.

NEW-AZHARDENEDBACKUP.PS1 · BASE TEMPLATE POWERSHELL

```
<#
.SYNOPSIS  Hardened Recovery Services vault + enhanced (hourly) VM policy + VM protection.
.NOTES     Requires Az.Accounts, Az.RecoveryServices, Az.Compute. Roles: Backup Contributor on the backup
RG,
           Virtual Machine Contributor on the workload RG. Set redundancy BEFORE protecting any item.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)] [string]   $SubscriptionId,
  [Parameter(Mandatory)] [string]   $ChangeRef,
  [Parameter(Mandatory)] [string[]] $VmNames,
  [Parameter(Mandatory)] [string]   $VmResourceGroup,
  [string] $ResourceGroup  = 'rg-backup-gwc',
  [string] $Location       = 'germanywestcentral',
  [string] $VaultName      = 'rsv-prod-gwc-01',
  [ValidateSet('LocallyRedundant','ZoneRedundant','GeoRedundant')] [string] $Redundancy =
'ZoneRedundant',
  [ValidateRange(14,180)] [int] $SoftDeleteDays = 28,
  [ValidateSet('Disabled','Unlocked','Locked')] [string] $Immutability = 'Unlocked',  # Locked is
irreversible
  [ValidateSet(4,6,8,12)] [int] $IntervalHours = 4,
  [int] $DailyRetentionDays = 14,
  [ValidateRange(1,30)] [int] $InstantRestoreDays = 7,
  [string] $CmkKeyUri                                    # optional:
https://<kv>.vault.azure.net/keys/<key>
)
$ErrorActionPreference = 'Stop'
Connect-AzAccount -Subscription $SubscriptionId | Out-Null
$tags = @{ ChangeRef = $ChangeRef; Owner = (Get-AzContext).Account.Id; Purpose = 'Backup' }

# ---- Stage 1: vault ----------------------------------------------------------
if ($PSCmdlet.ShouldProcess($VaultName, 'Create/harden Recovery Services vault')) {
  $vault = Get-AzRecoveryServicesVault -ResourceGroupName $ResourceGroup -Name $VaultName -ErrorAction
SilentlyContinue
  if (-not $vault) {
    $vault = New-AzRecoveryServicesVault -ResourceGroupName $ResourceGroup -Name $VaultName -Location
$Location -Tag $tags
  }
  # ---- Stage 2: redundancy (only possible while the vault is empty) ----------
  Set-AzRecoveryServicesBackupProperty -Vault $vault -BackupStorageRedundancy $Redundancy
  # ---- Stage 3: soft delete, immutability, private-only access ------------
  Set-AzRecoveryServicesVaultProperty -VaultId $vault.ID -SoftDeleteFeatureState Enable `
    -SoftDeleteRetentionPeriodInDays $SoftDeleteDays | Out-Null
  Update-AzRecoveryServicesVault -ResourceGroupName $ResourceGroup -Name $VaultName `
    -ImmutabilityState $Immutability -PublicNetworkAccess Disabled | Out-Null
  # ---- Stage 4 (optional): customer-managed key via system-assigned identity -
  if ($CmkKeyUri) {
    Update-AzRecoveryServicesVault -ResourceGroupName $ResourceGroup -Name $VaultName -IdentityType
SystemAssigned | Out-Null
```

```
    Write-Warning 'Grant the vault identity "Key Vault Crypto Service Encryption User" on the key, then
continue.'
    Set-AzRecoveryServicesVaultProperty -VaultId $vault.ID -EncryptionKeyId $CmkKeyUri `
      -KeyVaultSubscriptionId $SubscriptionId -UseSystemAssignedIdentity $true | Out-Null
  }
}

# ---- Stage 5: enhanced policy, every N hours, 24 h window ---------------------
$polName = "pol-vm-$($IntervalHours)h-$($DailyRetentionDays)d"
$pol = Get-AzRecoveryServicesBackupProtectionPolicy -VaultId $vault.ID -Name $polName -ErrorAction
SilentlyContinue
if (-not $pol -and $PSCmdlet.ShouldProcess($polName, 'New enhanced backup policy')) {
  $sch = Get-AzRecoveryServicesBackupSchedulePolicyObject -WorkloadType AzureVM -BackupManagementType
AzureVM `
          -PolicySubType Enhanced -ScheduleRunFrequency Hourly
  $sch.HourlySchedule.WindowStartTime = (Get-Date -Date '2026-01-01T00:00:00Z').ToUniversalTime()
  $sch.HourlySchedule.WindowDuration  = 24
  $sch.HourlySchedule.Interval        = $IntervalHours
  $ret = Get-AzRecoveryServicesBackupRetentionPolicyObject -WorkloadType AzureVM -ScheduleRunFrequency
Hourly
  $ret.DailySchedule.DurationCountInDays = $DailyRetentionDays
  $pol = New-AzRecoveryServicesBackupProtectionPolicy -Name $polName -WorkloadType AzureVM `
          -RetentionPolicy $ret -SchedulePolicy $sch -VaultId $vault.ID
  $pol.SnapshotRetentionInDays = $InstantRestoreDays
  Set-AzRecoveryServicesBackupProtectionPolicy -Policy $pol -VaultId $vault.ID
}

# ---- Stage 6: protect VMs -----------------------------------------------------
$protected = foreach ($vm in $VmNames) {
  if ($PSCmdlet.ShouldProcess($vm, "Enable-AzRecoveryServicesBackupProtection ($polName)")) {
    Enable-AzRecoveryServicesBackupProtection -ResourceGroupName $VmResourceGroup -Name $vm -Policy $pol
-VaultId $vault.ID | Out-Null
    [pscustomobject]@{ Vm = $vm; Policy = $polName }
  }
}

# ---- Evidence -----------------------------------------------------------------
[pscustomobject]@{
  ChangeRef = $ChangeRef; Operator = (Get-AzContext).Account.Id; TimeUtc = (Get-Date).ToUniversalTime()
  VaultId = $vault.ID; Redundancy = $Redundancy; SoftDeleteDays = $SoftDeleteDays; Immutability =
$Immutability
  Cmk = [bool]$CmkKeyUri; Policy = $polName; IntervalHours = $IntervalHours; Protected = $protected
} | ConvertTo-Json -Depth 4 | Out-File ".\$ChangeRef-backup-evidence.json"
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION — POWERSHELL

```
.\New-AzHardenedBackup.ps1 -SubscriptionId '<subscription-id>' -ChangeRef 'CHG0012345' `
  -VmResourceGroup 'rg-app-prod' -VmNames 'APPVM01','APPVM02' -Redundancy ZoneRedundant `
  -IntervalHours 4 -DailyRetentionDays 14 -Immutability Unlocked -WhatIf
```

4.1 SQL Server in a VM: workload policy hints

SQL databases are protected as *workload* items. The outline below shows the cmdlet sequence; property names on the schedule objects vary slightly between Az versions, so

inspect them with `Get-Member` before scripting.

SQL IN AZURE VM (OUTLINE) POWERSHELL

```powershell
$vault = Get-AzRecoveryServicesVault -ResourceGroupName 'rg-backup-gwc' -Name 'rsv-prod-gwc-01'
$sqlVm = Get-AzVM -ResourceGroupName 'rg-app-prod' -Name 'SQLVM01'
Register-AzRecoveryServicesBackupContainer -ResourceId $sqlVm.Id -BackupManagementType AzureWorkload `
  -WorkloadType MSSQL -VaultId $vault.ID -Force
$sch = Get-AzRecoveryServicesBackupSchedulePolicyObject -WorkloadType MSSQL      # full + diff + log
sections
$sch.IsLogBackupEnabled = $true
$sch.LogBackupSchedulePolicy.ScheduleFrequencyInMins = 15                        # sets Tier 1 RPO
$ret = Get-AzRecoveryServicesBackupRetentionPolicyObject -WorkloadType MSSQL
$pol = New-AzRecoveryServicesBackupProtectionPolicy -Name 'pol-sql-full-daily-log15' -WorkloadType MSSQL
`
         -SchedulePolicy $sch -RetentionPolicy $ret -VaultId $vault.ID
$db  = Get-AzRecoveryServicesBackupProtectableItem -WorkloadType MSSQL -ItemType SQLDataBase -VaultId
$vault.ID |
       Where-Object { $_.ServerName -like 'SQLVM01*' -and $_.Name -like '*ERPDB*' }
Enable-AzRecoveryServicesBackupProtection -ProtectableItem $db -Policy $pol -VaultId $vault.ID
```

CHAPTER 5

Methods C and D · Veeam Backup for Microsoft Azure and Site Recovery

5.1 Method C · Veeam Backup for Microsoft Azure

Veeam Backup for Microsoft Azure is deployed as an appliance VM from the Azure Marketplace. It takes snapshots of VMs, Azure SQL and Azure Files, writes backups into **repositories** (Azure Blob containers) in any region, and can be managed stand-alone or from Veeam Backup & Replication. Its strength in this design is the cross-region copy to West Europe and the portable backup format, which supports the EU Data Act's anti-lock-in goals.

1. **Deploy** the appliance from Marketplace into a management subnet in Frankfurt, without a public IP; reach its web console through Bastion or the P2S VPN. Protect the appliance itself with Azure Backup.
2. **Service account**: register an Entra application with the permissions Veeam documents for your version, scoped to the protected resource groups. Store the secret in Key Vault and rotate it.
3. **Repositories**: create a storage account in each region with public access disabled and private endpoints. Enable Blob versioning and an immutability policy for the West Europe container. Turn on Veeam repository encryption (AES-256) and store the password or key in Key Vault.
4. **Policy 1 · Standard VMs**: snapshots every 4 hours from 00:00 UTC, application-aware processing, local retention 14 days, backup copy to the West Europe repository with 35 days' retention.
5. **Policy 2 · SQL**: application-aware VM backups for SQL VMs. Let *one* engine own transaction-log backups (see the warning in Method A).
6. **Policy 3 · Tier 1**: the same 4-hour backups as Policy 1, combined with Site Recovery replication (Method D) for fast failover.
7. **Alerts**: send Veeam job notifications and appliance logs to the operations mailbox and Log Analytics; review the overnight report every morning.

> **NOTE · MENU NAMES CHANGE**
>
> Veeam reorganizes its console between versions. The concepts (service account, repositories, policies, backup copy, immutability, encryption) are stable; check the user guide for your release for exact screens.

5.2 Method D · Azure Site Recovery for Tier 1

Site Recovery continuously replicates the disks of Tier 1 VMs from Frankfurt to West Europe. Crash-consistent recovery points are created every few minutes and application-consistent points on a schedule you set, so RPO is seconds to minutes. A **recovery plan** groups VMs into boot order (database, then application, then web) with scripts for DNS or configuration changes.

1. Create a Recovery Services vault for DR *in the target region* (`rsv-dr-weu-01`), so it survives the loss of Frankfurt.
2. Prepare the DR VNet in West Europe with the same subnets and NSGs as production, but different address ranges if both will ever be connected.
3. In the VM blade select **Disaster recovery**, target region West Europe, choose the DR vault, replication policy (e.g. 24 h retention, app-consistent every 4 h) and target RG/VNet. Repeat per VM or enable from the vault.
4. Build a recovery plan with groups and manual actions, then run a **test failover** into an isolated test VNet. Clean it up afterwards.

SITE RECOVERY (OUTLINE OF THE AZ CMDLET SEQUENCE) POWERSHELL

```powershell
$dr = Get-AzRecoveryServicesVault -ResourceGroupName 'rg-dr-weu' -Name 'rsv-dr-weu-01'
Set-AzRecoveryServicesAsrVaultContext -Vault $dr
$priFab = New-AzRecoveryServicesAsrFabric -Azure -Location 'germanywestcentral' -Name 'fab-gwc'   # wait for job
$recFab = New-AzRecoveryServicesAsrFabric -Azure -Location 'westeurope'         -Name 'fab-weu'
$priPc  = New-AzRecoveryServicesAsrProtectionContainer -InputObject $priFab -Name 'pc-gwc'
$recPc  = New-AzRecoveryServicesAsrProtectionContainer -InputObject $recFab -Name 'pc-weu'
New-AzRecoveryServicesAsrPolicy -AzureToAzure -Name 'pol-asr-tier1' -RecoveryPointRetentionInHours 24 `
  -ApplicationConsistentSnapshotFrequencyInHours 4
# Then: New-AzRecoveryServicesAsrProtectionContainerMapping (gwc -> weu),
#       New-AzRecoveryServicesAsrAzureToAzureDiskReplicationConfig (per disk, with a cache storage account in Frankfurt),
#       New-AzRecoveryServicesAsrReplicationProtectedItem -AzureToAzure (per VM),
#       Start-AzRecoveryServicesAsrTestFailoverJob -Direction PrimaryToRecovery (isolated VNet) and cleanup.
```

WHY IT MATTERS · ASR IS NOT A BACKUP

Site Recovery keeps only short-term recovery points. Corruption or ransomware inside the VM replicates within minutes. Tier 1 systems therefore keep their 4-hour backups (Methods A–C) *and* replication.

CHAPTER 6

Method E · Restore Testing, Ransomware Recovery and Failover

6.1 Monthly restore test procedure

A backup that has never been restored is a hypothesis. Each month, restore at least one workload from each tier and measure the result.

1. **Select** one Tier 1, one Tier 2 and one Tier 3 workload (rotate so every workload is tested at least yearly) and a random recovery point within retention.
2. **Restore into isolation**: an isolated test VNet with no peering, using Vault › Backup items › Restore VM › Create new, a Veeam restore to a new VM, or an ASR test failover.
3. **SQL point-in-time**: restore a database as a new name to a random time (e.g. 13:47) and confirm the last committed transaction matches.
4. **Off-site path**: at least quarterly, restore from the West Europe copy, not the Frankfurt one.
5. **Measure** actual RPO (age of the restored data) and RTO (start to usable service) against the tier targets; record evidence and open improvement actions for any miss.
6. **Clean up** test resources and record the test in the change/test log.

RESTORE A VM RECOVERY POINT AS A NEW VM (TEST) — POWERSHELL

```
$vault = Get-AzRecoveryServicesVault -ResourceGroupName 'rg-backup-gwc' -Name 'rsv-prod-gwc-01'
$item  = Get-AzRecoveryServicesBackupItem -BackupManagementType AzureVM -WorkloadType AzureVM -VaultId
$vault.ID |
         Where-Object Name -like '*APPVM01*'
$rp    = Get-AzRecoveryServicesBackupRecoveryPoint -Item $item -VaultId $vault.ID | Get-Random
Restore-AzRecoveryServicesBackupItem -RecoveryPoint $rp -VaultId $vault.ID -TargetResourceGroupName 'rg-
restore-test' `
  -StorageAccountName 'strestorestaging01' -StorageAccountResourceGroupName 'rg-restore-test' `
  -TargetVMName 'APPVM01-RTEST' -TargetVNetName 'vnet-isolated-test' -TargetVNetResourceGroup 'rg-
restore-test' `
  -TargetSubnetName 'snet-test'
```

Evidence to keep	Example
Test record	Workload, tier, recovery point time, operator, start/end times, measured RPO/RTO, pass/fail.
Technical proof	Restore job IDs, screenshots of the application login, SQL query result at the chosen point in time.
Follow-up	Problem/change records for any target missed; sign-off by the service owner.

6.2 Ransomware or data corruption

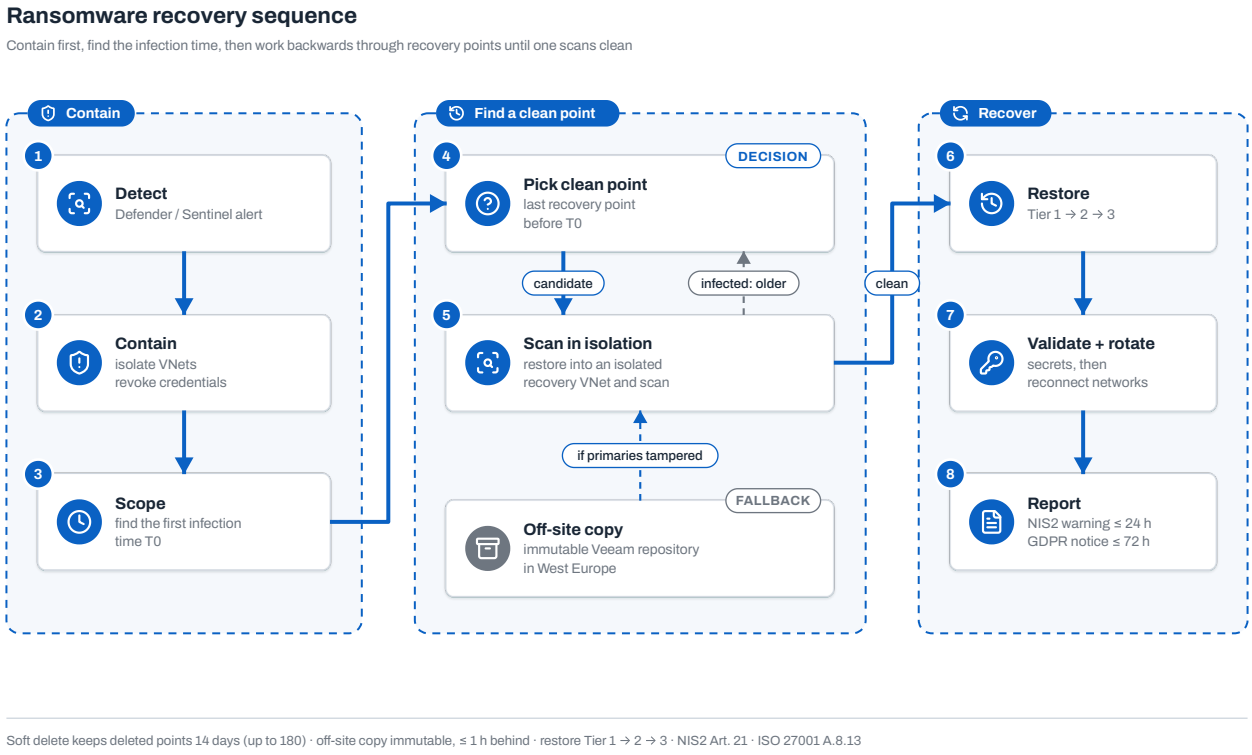


Figure 9. Never trust a recovery point until it has been scanned in isolation; working backwards from T0 is slower but avoids restoring the infection.

- Do not restore into the compromised network. Use an isolated recovery VNet and new credentials.
- Assume attackers had admin rights: verify that soft delete, immutability and MUA prevented tampering, and check vault activity logs.
- Scan restored disks with Microsoft Defender (or Veeam's malware scanning / Secure Restore when restoring through Veeam Backup & Replication) before reconnecting them.
- Restore in tier order and rotate every secret, key and service-account credential that existed before T0.

6.3 Regional failover

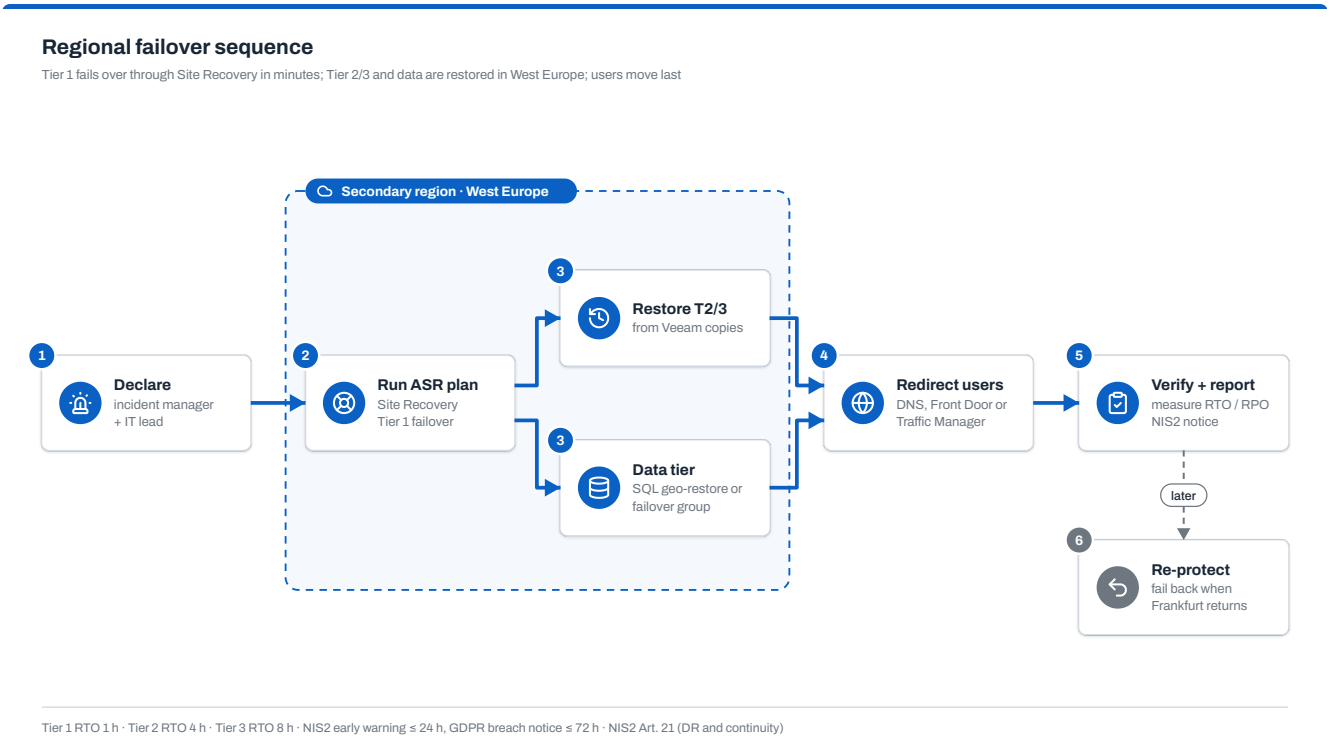


Figure 10. Users are redirected last, only after Tier 1 and the data tier are running in West Europe, so they never land on a half-restored service.

Declaring a disaster is a business decision, recorded with its time. Once Frankfurt is healthy again, **re-protect** the failed-over VMs (replicating West Europe back to Frankfurt), plan a quiet failback window, and repeat the verification steps.

CHAPTER 7

Regulatory Mapping, Cost and Troubleshooting

7.1 Why regulators care about backups

European rules rarely say "use product X". They require *outcomes*: availability can be restored in time, data stays where promised, incidents are reported, and management is accountable. The table teaches which design decision answers which expectation. It is not legal advice: classification (for example whether an operator is an essential or important entity, or meets KRITIS thresholds) must be confirmed with counsel and the authorities.

Framework	What it expects	How this design responds
GDPR / DSGVO	Art. 32: security of processing, including the ability to restore availability and access to personal data in a timely manner; Art. 33: breach notification within 72 hours.	Tested restores with measured RTO; encryption at rest (CMK) and in transit; EU-only regions; retention-based deletion after 35 days.
NIS2 and Germany's implementing act	Art. 21 risk measures explicitly include backup management, disaster recovery and business continuity; staged incident reporting (early warning within 24 hours); management accountability and training.	Tiered RPO/RTO, multi-region copies, immutable backups, alerting into Sentinel, quarterly management reporting.
KRITIS (if thresholds are met)	State-of-the-art security for critical services, regular audits, business continuity.	Documented DR plan, regular failover tests, MFA and PIM for all backup administration.
BSI C5	Cloud provider controls and transparency about data locations.	Azure's C5 attestation as the provider baseline; this design documents every backup location, key and role.
BSI IT-Grundschutz / ISO 27001	A data-backup concept (module CON.3); ISO 27001:2022 control A.8.13 Information backup.	This SOP plus the tier inventory and test records form the backup concept.
Seveso III	Safety management and emergency planning for sites with hazardous substances.	Systems supporting safety processes are Tier 1 with replication and short RTO.
EU Data Act (since Sept 2025)	Cloud switching and data portability without undue lock-in.	Copies in a portable third-party format and documented export paths.

7.2 Cost considerations

- **Protected-instance fees** per VM or SQL instance plus **backup storage** per GB; ZRS and GRS cost more than LRS.
- **Instant-restore snapshots** are billed as snapshot storage; 7 days is a good balance between speed and cost.
- **Cross-region transfer** (Veeam copies, ASR replication) is billed per GB; ASR adds a per-VM license fee and target storage.
- Veeam licensing, Key Vault operations and Log Analytics ingestion are smaller but recurring. Tier older Veeam backups to cool/archive Blob storage where restore time allows.

7.3 Troubleshooting

Symptom	Likely cause and fix
Cannot change storage redundancy	Items are already protected. Create a new vault with the right setting and move protection, or accept the current one.
VM backup fails with extension or snapshot errors	VM agent outdated or blocked outbound access; with private endpoints, check private DNS resolution of the vault endpoints.
SQL log backups fail or chain broken	Another tool also takes log backups. Choose one owner; others use copy-only.
Cannot reduce retention or delete data	Immutability or MUA is doing its job. Request Resource Guard access through the approval process.
CMK vault errors after key change	Vault identity lost *Crypto Service Encryption User* on the key, or the key was disabled; restore the permission or key version.
ASR replication health critical	Cache storage account or outbound connectivity to Site Recovery endpoints blocked; check NSG/firewall and service tags.
Restore test exceeds RTO	Restoring from vault tier instead of snapshot tier, undersized targets, or manual steps not scripted. Tune and retest.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- RPO and RTO per tier drive frequency, retention and tooling. Replication gives speed; backups give history. You need both.
- GRS only reaches the paired region; copies in a chosen second region need Veeam or Site Recovery.
- Harden vaults with soft delete, immutability, MUA, CMK and private endpoints, and separate backup, security and DR duties.
- Monthly isolated restore tests are what turn a design into evidence for GDPR, NIS2 and auditors.

8.2 Glossary

RPO / RTO	Maximum tolerable data loss / downtime.
LRS / ZRS / GRS	Redundancy within a datacenter / across zones / to the paired region.
Recovery Services vault	Vault for VM, SQL-in-VM, Files backup and Site Recovery.
Immutable vault	Vault that blocks early destruction of recovery points; Locked is irreversible.
MUA / Resource Guard	Second-party approval for critical backup operations.
CMK	Customer-managed key in Key Vault used to encrypt backup data.
Application-consistent	Copy taken after the application flushes writes.

Check your understanding

1. A database is replicated every 30 seconds to West Europe. Why is it still not protected against ransomware?
2. Why does a GRS vault in Germany West Central not satisfy a requirement for copies in the Netherlands?
3. Which three vault features stop an attacker with backup-admin rights from destroying recovery points?
4. Which SQL backup type determines the Tier 1 RPO, and how often does it run in this design?
5. Which roles would you give the daily backup operator, the security team and the vault's managed identity?